Jana Mostafa Sabry

Data Scientist | Machine Learning & AI Specialist | Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and Software Engineer with strong expertise in Machine Learning, Deep Learning, Natural Language Processing (NLP), and Full-Stack Web Application Development. Proven track record in developing predictive models, unsupervised anomaly detection pipelines, fine-tuning Large Language Models (LLMs), and building end-to-end cloud and mobile solutions. Adept at turning complex datasets into actionable business intelligence and scalable software applications.

EDUCATION

The British University in Egypt (BUE) – Faculty of Informatics and Computer Science *Sep 2022 – Jun 2026*
Cairo, Egypt

Digital Egypt Pioneers Initiative (DEPI) – Data Science Track *Jun 2025 – Dec 2025*
6-Month Scholarship Program

SKILLS

- **Technical Focus:** Data Science, Data Analysis, Machine Learning, NLP, Deep Learning, Object Detection, Web Development, FlutterFlow, Operating Systems, Data Structures
- **Programming Languages:** Python, C#, Java, SQL, JavaScript, Dart, C
- **Tools & Platforms:** HTML, CSS, MySQL, Microsoft Word, Microsoft PowerPoint, Microsoft Excel
- **Soft Skills:** Communication, Leadership, Problem-Solving, Public Speaking

EXPERIENCE

Software Engineer (Founding Technology Team) — Osta

Egyptian home-services marketplace connecting homeowners with vetted craftsmen through open bidding | Next.js, TypeScript, React, Supabase (PostgreSQL), PWA

- Owned the bilingual customer web app (Arabic RTL / English LTR with next-intl) and its API endpoints: job posting with photos and video, live quote comparison, craftsman selection, job history and two-way ratings.
- Co-owned the admin panel: craftsman KYC verification queue, support-ticket and job overview dashboards, and a read-only audit-log viewer behind mandatory TOTP multi-factor login.
- Co-designed the shared Supabase/PostgreSQL schema and API contract with row-level security on every table, an atomic row-locked quote-selection transaction, and a hashed one-time arrival passcode.
- Verified the core marketplace loop (post, bid, select, arrive, complete) end to end against a live database, backed by a Vitest suite of 930+ passing tests, strict TypeScript and clean lint.

Machine Learning Intern — FlyRank

Research paper: janamoustafa.github.io/FlyRank-ML | Jul – Sep 2026 | Python, Scikit-learn, Pandas, Random Forest

- Built an end-to-end CTR opportunity-scoring model that ranks which web pages an SEO content team should rewrite first, trained on 22,006 URLs from 30 enterprise clients using Google Search Console and GA4 data.
- Reached 0.922 ROC-AUC under five-fold GroupKFold validation on unseen clients and 65.6% precision in the top 100 pages, a 6.56× lift over the 10% base rate and 20.6 points above a logistic-regression benchmark.
- Audited the data for leakage, removing forward-looking and target-proxy columns, wrote a data contract, and limited the model to five features available at decision time.

- Turned the scores into an action playbook with reason codes, a 50-page sprint plan worth an estimated 6,184 extra clicks, human-review safeguards and drift-based retraining triggers, published as a research paper.

Full-Stack Web Developer — Fridge Check (Personal Project)

Live app: fridge-check-sooty.vercel.app | Next.js, React, TheMealDB API, Vercel

- Designed, built and deployed a bilingual (English / Arabic) recipe-finder web app that matches recipes to the ingredients a user already has and lists exactly what is missing.
- Implemented free-text and quick-add ingredient entry, diet filters (vegetarian, vegan, gluten-free, dairy-free, pescatarian), best-match and fewest-missing sorting, and a pantry-staples option.
- Added saved recipes and an auto-generated shopping list, integrated the TheMealDB API for recipe data and photos, and shipped it as an installable PWA with continuous deployment on Vercel.

Full-Stack Web Developer — Saving Goose (Personal Project)

Live app: saving-goose.vercel.app | HTML, CSS, JavaScript, PWA, Web Push, Vercel

- Built and deployed a bilingual (English / Arabic) privacy-first budgeting PWA with no accounts or ads: a monthly dashboard showing spent, remaining and safe-to-spend amounts, spending trends and all-time history.
- Implemented payment logging with categories, recurring payments that post themselves on their due day, savings goals with streak tracking, and a guided setup that builds a personal money plan.
- Added hourly check-in reminders delivered by a push-notification server while the app is closed, optional motivational lines, backup export and restore, and CSV export, with all financial data kept on the device.

Machine Learning Intern — Nokia

- Engineered a robust machine learning pipeline using Scikit-learn to predict network KPIs, achieving an $R^2 > 0.99$ for throughput prediction using XGBoost, LightGBM, and Random Forest.
- Developed an automated data preprocessing workflow featuring cyclical time encoding, Haversine distance calculations, and rolling time-series statistics to capture complex spatial-temporal patterns.
- Implemented an unsupervised anomaly detection strategy using Isolation Forest to generate target labels from raw datasets, enabling downstream supervised classification.

Technical Intern — Multiple Organizations

Uneeq Interns | Saiket Software Company | National Telecommunications Institute (NTI) | Information Technology Institute (ITI)

- Cleaned and analyzed datasets, created visualizations and insights, and trained ML/deep learning models across multiple data science projects.
- Built a full-stack interactive application for meal planning using Dart and FlutterFlow, covering both front-end and back-end.
- Trained and deployed object detection models across various domains, including data preprocessing and cleaning.
- Designed a server-client communication system enabling multi-entity messaging and notifications using the RMI concept.
- Developed full-stack interactive games using Java and Python.
- Built a database management system with query-based information extraction.
- Designed and developed a responsive personal portfolio website using HTML and CSS.

SELECTED PROJECTS

- **Osta –** Egyptian home-services marketplace with open bidding between homeowners and vetted craftsmen; owned the bilingual customer app and co-owned the admin panel and shared Supabase/PostgreSQL schema (Next.js, TypeScript, React, Supabase, PWA).
- **Fridge Check –** Bilingual recipe-finder PWA that matches recipes to the ingredients you already have and lists what is missing, with diet filters, saved recipes and a shopping list (Next.js, React, TheMealDB API, Vercel) — fridge-check-sooty.vercel.app

- **Saving Goose –** Bilingual, privacy-first budgeting PWA with a monthly budget dashboard, recurring payments, savings goals and streaks, push-notification reminders, and backup/CSV export (local-first data, Vercel) — saving-goose.vercel.app
- **Client Portfolio & Company Websites –** Designed and built responsive personal portfolio sites for two clients with HTML, CSS and JavaScript (hana-portfolio-peach.vercel.app, hana-portfolioo.vercel.app) and the company website for AOM Tech, an AI solutions startup, with Next.js and React, all deployed on Vercel (aom-tech.vercel.app).
- **Bone Tumor Detection System –** Deep learning system for tumor detection using models trained from scratch alongside fine-tuned pretrained models.
- **NLP & LLM Fine-Tuning Projects –** Implemented multiple NLP projects leveraging large language models (LLMs) and various fine-tuning methods.
- **Cloud Storage System –** Full-stack system (backend, frontend, and DevOps) featuring encryption, database linking, and multiple security measures.
- **Multi-Agent Communication System –** Designed and implemented communication and coordination logic between multiple software agents.
- **Emotion Detection System –** Advanced machine learning project for emotion recognition and classification.
- **Traffic Light Manager –** Hardware-based traffic light control system built by manipulating circuit components.
- **Tetris Game –** Interactive, responsive Tetris game developed using C.
- **Console-Based Calculator –** Command-line calculator performing basic arithmetic operations, built in C.

CERTIFICATIONS

IBM – Deep Learning Engineering Specialization *Jul 20, 2024*
Certificate of Competency in Deep Learning Course.
[Click here to view Certificate]

IBM – Data Analysis with Python *Jul 27, 2024*
Certificate of Competency in Data Analysis with Python Course.
[Click here to view Certificate]

Impact – Cyber Security *Aug 2024*
Certificate of Achievement, sponsored by the British university in Egypt.
[Click here to view Certificate]

International Accreditation Organization (IAO) – Cyber Security *Nov 2024*
Certificate of Achievement, sponsored by International Accreditation Organization (IAO) and Impact.
[Click here to view Certificate]

IBM – Machine Learning with Python *Dec 23, 2024*
Certificate of Competency in Machine Learning with Python Course.
[Click here to view Certificate]

IBM – IBM Data Science Professional Certificate *Jan 1, 2025*
Certificate of Competency Data Science Professional Course.
[Click here to view Certificate]

ITI – Artificial Intelligence *Jul 2025*
Certificate of Competency (90 hrs.) in Artificial Intelligence fundamentals.
[Click here to view Certificate]

Digital Egypt Pioneers Program (DEPI) – Data Scientist *Jun – Dec 2025*
Certificate of Achievement, sponsored by the Ministry of Communications and Information Technology (MCIT) of Egypt.
[Click here to view Certificate]

UneeQ – Machine Learning Internship *Aug 2025*
Certificate of Competency in Machine Learning Internship.
[Click here to view Certificate]

Nokia – Machine Learning Internship *Aug 2025*
Certificate of Active Participation in Nokia MN Summer Internship 2025_Egypt.
[Click here to view Certificate]

National Telecommunications Institute (NTI) – AI for Business *Aug 2025*
Certificate of Competency in Summer training course of AI for Business.
[Click here to view Certificate]

NVIDIA – Getting Started with Deep Learning *Aug 29, 2025*
Certificate of Competency in Deep Learning fundamentals.
[Click here to view Certificate]

SaiKet Systems Software Company – Machine Learning Internship *Sep 2025*
Certificate of Competency in Machine Learning Internship.
[Click here to view Certificate]

FlyRank – Machine Learning Internship *Sep 9, 2026*
Certificate of Completion, Machine Learning Internship Program (Jul – Sep 2026).
[Click here to view Certificate]

LANGUAGES

- Arabic — Native or Bilingual Proficiency
- English — Fluent